

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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Anonymous submission

ABSTRACT

Counterfactual explanations (CE) prescribe how targeted input changes would alter a model’s prediction, yet in practice they routinely perturb far more features than the target effect requires, diluting their interpretability and eroding their actionability. We introduce *COLA*, a model- and generator-agnostic post-hoc framework that refines any given CE by first computing an optimal-transport (OT) coupling between the factual and counterfactual sets and then using that coupling to drive a Shapley-based attribution, *p-SHAP*, which isolates a minimal set of edits while exactly preserving the target effect. This coupling-informed formulation unifies the widely used Shapley variants as special cases of a single attribution scheme and recasts feature attribution as a transport problem that minimizes the cost of the explanation. Theoretically, we prove that the OT coupling minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and that, under a mild Lipschitz condition satisfied by the diverse model classes we evaluate, the refined counterfactuals provably never move farther from the factuals than the originals. Empirically, across four datasets, twelve models, and five CE generators—spanning instance-level, group-level, and distributional counterfactuals—COLA attains the identical target effects while retaining only 26–45% of the original feature edits, and on a tractable small-scale benchmark it closely tracks the global optimum computed by mixed-integer programming. Together, these results establish COLA as a general and dependable route to sparse, actionable counterfactuals.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) is foundational to the transparency and trustworthiness of modern artificial intelligence systems (Arrieta et al., 2020; Das & Rad, 2020). Within this area, feature attributions (FA) methods—most prominently Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017)—quantify the contribution of each input feature to a machine learning (ML) model’s output, distilling a complex decision function into its most influential drivers. In a healthcare model, for instance, Shapley values surface decisive factors such as age and medical history, thereby helping clinicians interpret the model’s decisions (Ter-Minassian et al., 2023; Nohara et al., 2022). A complementary paradigm, counterfactual explanations (CE) (Wachter et al., 2017; Guidotti, 2022), instead characterizes how targeted changes to the input would yield a different outcome. Although hundreds of CE algorithms have been proposed (Guidotti, 2022; Verma et al., 2020), no single algorithm serves every use case, precisely because each is tailored to a distinct scenario, objective, and level of granularity. One algorithm may seek a single counterfactual instance for each factual instance; another may treat multiple instances as a group and return one or several counterfactuals per factual observation. Some methods operate over the entire dataset, identifying global CE that prescribe the direction in which factual instances should move to attain the desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024); others regard the factual instances as an empirical distribution and seek a counterfactual distribution that preserves the shape of the factual distribution at comparable cost (You et al., 2025). The methods diverge further in their modeling assumptions, with some presupposing differentiable models and others targeting tree-based or ensemble models specifically. It is exactly this heterogeneity of goals, granularities, and model assumptions that a genuinely general refinement framework must accommodate—and that motivates

the generator- and model-agnostic design we develop, which is engineered to operate uniformly across this entire spectrum.

Problem Description and Challenges Although FA and CE are each indispensable for rendering AI models interpretable and accountable, neither is sufficient in isolation: FA identifies influential features but rarely prescribes tangible, actionable steps, whereas CE prescribes concrete changes yet typically perturbs more features than the target effect requires, yielding recommendations that are impractical to enact. We therefore formulate a single, general problem that unifies the two paradigms and consolidates the extensive foundations established across the literature (see Appendix A). Concretely, we set out to answer the following question:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
$\textcircled{a}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{a}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{a}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{R}$
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform seeks to increase user registrations. It has recorded user interactions—the amount of money spent ($\textcircled{a}$), the number of clicks ($\blacktriangleright$), and whether the user has registered ($\textcircled{R}$). In the original data ($\mathbf{x}$), no user is registered. The two action plans $\mathbf{z}'$ and $\mathbf{z}''$ each adjust user characteristics to attain the desired outcome ($\mathbf{y}^*$) of full registration. Although both realize an identical (half) counterfactual effect, $\mathbf{z}''$ achieves it with strictly fewer feature modifications than $\mathbf{z}'$, thereby instantiating the modification-minimality objective that is the central concern of this work.

Three principal challenges render this problem demanding, and each dictates a corresponding design requirement. First, because no single CE algorithm can meet all needs and the problem is inherently task-specific, a viable solution must be agnostic to the choice of generator. Second, to remain applicable across the broadest range of models, the approach must dispense with strong assumptions about f —such as differentiability or a particular architectural structure—and instead treat the model strictly as a black box. Third, generic FA such as feature importance is actively misleading in this setting, because attribution scores computed in isolation are incoherent with the feature changes that actually drive the counterfactual effect; consequently, it is ineffective to perform FA independently of CE to select the most important features to change. We demonstrate that this decoupling provably induces counterproductive feature modifications (equivalently, the actions taken by users or stakeholders) in Result II of Section 6, since the features deemed important in general need not coincide with the specific pathways that achieve the desired counterfactual outcome. Taken together, these three challenges directly motivate the coupled, model-agnostic design we develop next, in which attribution and the counterfactual objective are bound together rather than treated in isolation.

Main Contributions This paper makes four coordinated contributions. (i) *A general post-hoc framework, COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)*. We introduce COLA, which refines any given CE—for arbitrary models and CE generators alike—by using an optimal transport (OT)-induced coupling between the factual and counterfactual sets to guide Shapley attribution, thereby binding attribution directly to the counterfactual objective rather than treating the two in isolation. (ii) *A new attribution method, p-SHAP*. The coupling gives rise to p -SHAP, a joint-probability-informed Shapley method that subsumes the commonly used Shapley variants as special cases under appropriate couplings and, through its modular interface for attribution and edit selection, exposes the factual-counterfactual alignment as an explicit, tunable design choice. (iii) *Theoretical guarantees*. We prove that the OT coupling minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and that, under a mild Lipschitz condition, the refined counterfactuals are provably no farther from the factuals than the originals. (iv) *Comprehensive empirical evidence*. Across four datasets, twelve models, and five CE methods—covering instance-level, group-level, and distributional counterfactuals—COLA with p -SHAP attains the same target effects while retaining only 26–45% of the original feature edits, and on a tractable small-scale benchmark it closely tracks

the mixed-integer-programming optimum. Together, contributions (iii) and (iv) show that the cost guarantee established in theory is realized consistently in practice across the full heterogeneity of models and generators.

To the best of our knowledge, COLA is the first method to address the problem of Figure 1 systematically *without committing to any particular CE algorithm or model class*, and it attains this generality while furnishing both provable guarantees and consistent empirical gains.

2 PROBLEM FORMULATION

We now formalize the problem illustrated in Figure 1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model, let $\mathbf{x}$ collect the observed (factual) data over n instances and d features ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, $d \geq 1$), and let $\mathbf{y}^*$ denote the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The task is to determine a (group of) counterfactual instance(s) $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) that, subject to a budget of at most C feature changes, drives the model output(s) as close as possible to $\mathbf{y}^*$. Writing D for a divergence that quantifies the dissimilarity between two entities, we state the problem as follows.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) together with the constraint equation (1b) constitutes the canonical CE optimization: $\mathbf{z}$ must bring $f(\mathbf{z})$ close to $\mathbf{y}^*$ while remaining proximate to $\mathbf{x}$, so that $\mathbf{z}$ serves as a contrastive reference explaining why $f(\mathbf{x})$ falls short of $\mathbf{y}^*$. We deliberately impose no restriction on the divergence D , a choice that preserves the full generality of the formulation; admissible instantiations include the Euclidean distance, OT, maximum mean discrepancy (MMD), and mean- or median-based differences of the model outcome. The remaining constraints equation (1c)–equation (1e) augment the CE optimization with an explicit action budget. Beyond the counterfactual data $\mathbf{z}$, we jointly optimize a binary indicator $\mathbf{c}$ that governs which entries may move: entry z_{ik} is permitted to change if and only if $c_{ik} = 1$, and constraint equation (1c) caps the total number of such changes at C . Inspecting equation (1d) and equation (1e), when $c_{ik} = 0$ the two bounds coincide and force $z_{ik} = x_{ik}$, holding position (i, k) fixed; when $c_{ik} = 1$, the sufficiently large constant M relaxes both bounds, granting z_{ik} full freedom to change. This binary indicator therefore encodes precisely the action plan, and its sparsity is exactly the modification-minimality objective of Figure 1.

To solve equation (1), we leverage FA to pinpoint the most influential features and thereby determine the modification indicator $\mathbf{c}$. The next section reviews the commonly used Shapley value methods for FA that, together with our proposed p -SHAP, are integrated into the algorithmic framework COLA to produce the refined counterfactual $\mathbf{z}$. The difficulty of this endeavor should not be understated: the problem is computationally hard, remaining $\mathcal{NP}$ -hard even in the degenerate regime in which $d = 1$ and f is linear, as we establish in Appendix B. This hardness result serves a dual purpose—it justifies the pursuit of a principled approximate framework and, at the same time, delineates the boundary within which exact optimality can realistically be sought—and thereby positions COLA as a polynomial-time refinement precisely where exact combinatorial solvers become intractable.

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews the commonly used Shapley value methods for FA that, together with our proposed p -SHAP, are integrated into the algorithmic framework COLA. We first recall the game-theoretic notion of the Shapley value and then survey the Shapley variants most widely used for FA, situating them along the single axis—the treatment of the factual–counterfactual alignment—along which p -SHAP will subsequently unify them; Sundararajan & Najmi (2020) provide a comprehensive overview.

In cooperative game theory, a coalitional game is specified by a finite set of players—here, the features—denoted $\mathcal{F} = \{1, 2, \dots, d\}$, together with a characteristic function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ that

assigns to each coalition $S \subseteq \mathcal{F}$ the total payoff $v(S)$ its members can attain through cooperation, subject to the normalization $v(\emptyset) = 0$. The Shapley value, defined below in terms of v , is the canonical solution concept in this framework, furnishing the unique axiomatically equitable allocation of the grand coalition’s payoff $v(\mathcal{F})$ among the individual players.

Before defining the Shapley value, let $\Delta(k, S)$ denote the marginal contribution of a player (feature) k to a coalition S —that is, the incremental payoff the coalition realizes upon admitting k :

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i of a player i is then defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k thus averages the normalized marginal contributions $\Delta(k, S)$ of feature k over all coalitions S that exclude k , weighting each contribution by the inverse binomial coefficient $\binom{d-1}{|S|}^{-1}$ and normalizing the outer sum by the feature count d . This construction yields the unique axiomatic allocation of the total payoff among features, with ϕ_k quantifying the contribution of feature k relative to the entire feature collection.

For an arbitrary data point $\mathbf{x}_i$, we write $\mathbf{x}_{i,S}$ for the sub-vector restricted to the features in S and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ for its complement, adopting the same convention throughout for the remaining notation. The Shapley variants used for FA differ chiefly in how they model the “absence” of a feature—equivalently, in the reference against which the marginal contribution is measured—and it is precisely this modeling choice that governs their compatibility with the counterfactual setting.

Baseline Shapley (B-SHAP). This method fixes a single baseline point $\mathbf{r}_j$ as the reference for any $\mathbf{x}_i$ and defines the set function

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

so that the absence of a feature is modeled by substituting the corresponding value from the reference point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). B-SHAP thereby presupposes an exact, deterministic alignment between $\mathbf{x}$ and $\mathbf{r}$, which is the most restrictive assumption among the three variants.

Random Baseline Shapley (RB-SHAP). RB-SHAP, a variant of B-SHAP employed implicitly in Lundberg & Lee (2017) and in the widely adopted “SHAP” library, relaxes the fixed baseline to an expectation over a background distribution:

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where the background distribution $\mathcal{D}$ is commonly taken to be the training set (Lundberg & Lee, 2017; Merrick & Taly, 2020). RB-SHAP thus dispenses with any explicit alignment, at the cost of ignoring counterfactual information entirely.

Counterfactual Shapley (CF-SHAP). CF-SHAP extends RB-SHAP by instantiating $\mathcal{D}$ as the counterfactual distribution conditioned on each individual factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$):

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

Adopted by Albini et al. (2022) and Kommiya Mothilal et al. (2021) with demonstrated advantages for contrastive FA, CF-SHAP presupposes a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$, and hence occupies an intermediate position between the deterministic alignment of B-SHAP and the alignment-free regime of RB-SHAP. These three variants thus trace a single progression from deterministic, through probabilistic, to absent alignment—the very axis that p -SHAP unifies.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize the entire family spanning equation (4) through equation (6) by introducing an algorithm A_{Prob} that returns the joint probability of the factual and counterfactual sets,

thereby recovering all three baselines as special cases of a single attribution scheme and elevating the factual–counterfactual alignment from an implicit modeling assumption to an explicit, tunable design choice.¹ Our p -SHAP is defined as follows,

$$\begin{aligned} v^{(i)}(S) &= \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_i, S; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \\ \text{s.t. } p &= A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \end{aligned} \quad (7a) \quad (7b)$$

Crucially, and in contrast to prior methods that rely on random baselines, p -SHAP confronts the alignment problem directly, identifying the optimal coupling via OT. This recasts feature attribution not as a mere decomposition of the prediction but as a transport problem that minimizes the cost of the explanation—a reformulation that is the conceptual pivot of our approach.

Two properties distinguish p -SHAP and account for its advantages. First, A_{Prob} is independent of the CE algorithm and depends only on the generated counterfactual $\mathbf{r}$. By conditioning solely on the factual and counterfactual sets rather than on the mechanism that produced them, p -SHAP delivers FA that is consistent across, and insulated from the variability of, disparate CE generation processes—a decisive advantage over CF-SHAP. Second, and contrary to common expectation, we show that OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by [Albini et al. \(2022\)](#), in Result II of Section 6 later: an inferred optimal coupling can constitute a better attribution reference than even the alignment dictated by the generator itself.

The central instantiation studied in this paper adopts the OT problem (equivalently, the 2-Wasserstein divergence) defined below, and takes the resulting transportation plan $\mathbf{p}_{\text{OT}}$ as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ within p -SHAP. This instantiation is the one on which our theoretical guarantees and empirical evaluation both center.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ denote the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, respectively, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of admissible joint distributions—that is, all transport plans consistent with those prescribed marginals. The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is an entropic regularizer with coefficient $\varepsilon \geq 0$; activating it ($\varepsilon > 0$) renders the objective strictly convex and, through the Sinkhorn iteration, substantially accelerates the computation of OT, whereas setting $\varepsilon = 0$ recovers the exact, unregularized transport plan on which our theoretical analysis rests. The regularizer thus supplies a tunable trade-off between computational efficiency and exactness without altering the underlying coupling objective.

Theoretical Aspects of p -SHAP Under the alignment it induces between the factual $\mathbf{x}$ and the counterfactual $\mathbf{r}$, OT minimizes the total cost of transporting $\mathbf{x}$ towards $\mathbf{r}$, which is precisely the objective of achieving the counterfactual outcome at minimal cost. We render this correspondence rigorous under a Lipschitz-continuity assumption on the predictive model f —an assumption sufficiently weak to encompass the full range of model classes we evaluate, from linear and kernel methods to tree ensembles and deep networks. Theorem 4.1 below (proved in Appendix C) establishes that the transportation plan $\mathbf{p}_{\text{OT}}$ employed by p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$: the 1-Wasserstein distance $W_1(f(\mathbf{x}), \mathbf{y}^*)$ is bounded by the Lipschitz constant of f times the square root of the minimized expected modification cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, in which the couplings p_{ij} ($j = 1, 2, \dots, m$) prescribe how the feature values of $\mathbf{x}_i$ should be adjusted towards those of one or more $\mathbf{r}_j$. Consequently, the OT plan $\mathbf{p}_{\text{OT}}$ within p -SHAP furnishes a modification strategy that steers the feature values of $\mathbf{x}$ towards those of $\mathbf{r}$ at the smallest attainable expected cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$. Because $\mathbf{p}_{\text{OT}}$ attains the minimal cost among all feasible plans $\mathbf{p} \in \Pi$, it simultaneously yields the tightest cost-proportional upper bound on the residual counterfactual violation $W_1(f(\mathbf{x}), \mathbf{y}^*)$; minimizing transport cost and minimizing the guaranteed counterfactual gap therefore coincide as a single objective.

¹First, p -SHAP reduces to B-SHAP in equation (4) when A_{Prob} specifies a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates a deterministic $i \leftrightarrow j$ alignment for any $\mathbf{x}_i, \mathbf{r}_j$. Second, it reduces to RB-SHAP in equation (5) when A_{Prob} is independent of CE and instead associates $\mathbf{x}$ with an arbitrary background distribution $\mathcal{D}$. Third, it reduces to CF-SHAP in equation (6) when A_{Prob} is built upon a known CE distribution.

Theorem 4.1 (*p*-SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 bounds the transport cost, it serves as a tractable convex proxy for the $\mathcal{NP}$ -hard discrete sparsity (L_0) problem. By concentrating attribution mass on the most cost-efficient transport paths, the OT coupling guides the greedy selection in Algorithm 1 to discard non-essential features. Beyond this cost guarantee, *p*-SHAP is conceptually sound in the manner in which it attributes model behavior to feature modifications: Theorem 4.2 below (proved in Appendix D) establishes that its value function admits a precise interventional interpretation, thereby elevating the attribution from a heuristic score to a well-defined interventional quantity.

Theorem 4.2 (Interventional Effect of *p*-SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Finally, *p*-SHAP inherits the desirable axioms of B-SHAP and RB-SHAP, which underwrites its soundness as a feature-attribution tool; we omit the proof and refer the reader to Sundararajan & Najmi (2020) and Lundberg & Lee (2017) for the corresponding Shapley axioms. Thus *p*-SHAP unites these established axiomatic guarantees with the cost-optimality and interventional interpretation derived above, so that its attributions are at once axiomatically principled, provably cost-efficient, and causally interpretable.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The algorithmic framework COLA, stated in Algorithm 1, solves equation (1) through four modular components executed in sequence: a CE generator A_{CE} , a coupling estimator A_{Prob} , a Shapley attributor A_{Shap} , and a value assigner A_{Value} . This modular decomposition is precisely what renders COLA simultaneously generator-agnostic and model-agnostic: each component may be instantiated independently, and only A_{Prob} and A_{Shap} embody the coupling-informed attribution that constitutes our methodological contribution. We detail Algorithm 1 below, in tandem with the illustration in Figure 2.

Line 1 (*Generating a counterfactual $\mathbf{r}$ via a CE algorithm*). The CE algorithm A_{CE} receives the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ , and returns a counterfactual $\mathbf{r}$ that remains close to $\mathbf{x}$, satisfying $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$. Because COLA treats A_{CE} strictly as a black box, any of the instance-level, group-level, or distributional generators of Section 6 may be substituted without modification—exactly the property that endows the framework with its generator-agnostic character.

Line 2 (*Estimating the joint distribution of $\mathbf{x}$ and $\mathbf{r}$*). The algorithm A_{Prob} maps the inputs $\mathbf{x}$ and $\mathbf{r}$ to a matrix encoding the joint distribution over the n and m data points of $\mathbf{x}$ and $\mathbf{r}$, respectively. This joint distribution $\mathbf{p}$ expresses an alignment (matching) between the factual and counterfactual rows and is subsequently consumed in Line 5 to compose the reference values for $\mathbf{z}$. As discussed in Section 3, when the row-level alignment is unknown—the prevailing case in practice— A_{Prob} is instantiated with OT to recover the joint distribution attaining the smallest OT distance between $\mathbf{x}$ and $\mathbf{r}$; when a faithful alignment is instead available, A_{Prob} adopts a joint distribution that reflects the

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

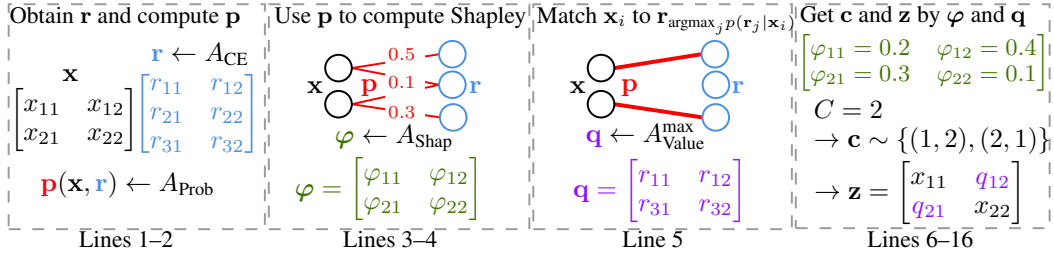


Figure 2: [An illustration of COLA] The figure traces how COLA produces the action plan $\mathbf{c}$ and the refined counterfactual $\mathbf{z}$ for equation (1) by executing the four sequential components A_{CE} , A_{Prob} , A_{Shap} , and A_{Value} in order. For clarity, line 5 is illustrated with the simpler value assigner $A_{\text{Value}}^{\max}$. In lines 6–16 we set $C = 2$, so that the sampling selects exactly two positions for modification according to the normalized attribution matrix φ .

row-level alignment of $\mathbf{x}$ and $\mathbf{r}$ directly. This single component thereby interpolates continuously between the alignment regimes that distinguish the Shapley variants of Section 3.

Lines 3–4 (*p-SHAP FA*). We evaluate equation (7) to compute the Shapley values of $\mathbf{x}$. The joint distribution $\mathbf{p}$ may be used here—though it is not strictly required—to align each factual row of $\mathbf{x}$ with its counterfactual rows in $\mathbf{r}$, so that the selected rows of $\mathbf{r}$ serve as the most representative contrastive reference for that factual row; as our ablations in Section 6 confirm, this alignment exerts a decisive influence on the quality of the contrastive FA. The resulting Shapley matrix is taken element-wise in absolute value and normalized to sum to one, yielding the attribution matrix φ , which thereby constitutes a proper probability distribution over feature positions.

Line 5 (*Assigning feature values*). The algorithm A_{Value} takes the counterfactual $\mathbf{r}$ and the joint distribution $\mathbf{p}$ as input and, for each factual row i , selects one or several rows of $\mathbf{r}$ to serve as references for modifying $\mathbf{x}$. It returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$ in which each entry q_{ik} is the counterfactual candidate value for x_{ik} ($\forall i, k$). We instantiate A_{Value} in two complementary forms, $A_{\text{Value}}^{\max}$ and $A_{\text{Value}}^{\text{avg}}$, which respectively select a single reference row and aggregate multiple reference rows, and which trade off sharpness of the reference against robustness to coupling uncertainty.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\max}$ selects the single row of $\mathbf{r}$ carrying the highest coupling probability:

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i), k} \text{ and } \tau(i) = \arg \max_{j=1, 2, \dots, m} p_{ij}. \quad (9)$$

The algorithm A^{avg} instead forms each q_{ik} as a convex combination—a coupling-weighted average—of $r_{1k}, r_{2k}, \dots, r_{mk}$, thereby pooling evidence across all counterfactual rows in proportion to their coupling mass:

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Because the non-negative matrix φ is normalized to sum to one, it constitutes a proper sampling distribution over positions; Line 7 accordingly treats it as a selection policy, drawing C positions of $\mathbf{x}$ for value replacement (i.e. setting $c_{ik} = 1$) with probability proportional to attributed importance. For every (i, k) with $c_{ik} = 1$, the value x_{ik} is overwritten by q_{ik} , and the resulting matrix is returned as $\mathbf{z}$, which together with $\mathbf{c}$ constitutes the optimized solution of equation (1).

By Theorem 4.1, COLA is designed to minimize the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by modifying $\mathbf{z}$ based on feature attribution results, which identify the most important features to adjust to achieve the desired counterfactual effect. The theorem below (proved in Appendix E) further guarantees that the refined $\mathbf{z}$ produced by COLA respects the proximity constraint equation (1b): measured in the Frobenius norm, $\mathbf{z}$ provably never strays farther from the factual $\mathbf{x}$ than the aligned counterfactual proposal does. The refinement step therefore cannot degrade proximity, and any reduction in the number of edits is thus obtained at no cost to closeness—a guarantee that Table 3 confirms holds broadly across datasets, models, and generators.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the complexity of A_{CE} , and, for A_{Shap} , suppose Shapley values are estimated by weighted linear regression over M_{Shap} sampled subsets. The overall complexity of COLA in the instance count n , the counterfactual count m , the feature dimension d , and the entropic-OT regularization parameter ε is then $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ when $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ when $A_{\text{Value}}^{\text{avg}}$ is used, with the full derivation deferred to Appendix F. Every term is polynomial, so COLA imposes only lightweight post-hoc overhead atop the chosen CE generator—in pointed contrast to the exponential blow-up of exact combinatorial solvers, and fully consistent with the $\mathcal{NP}$ -hardness of the underlying problem. COLA therefore delivers its sparsity gains at a computational cost that scales gracefully with the problem dimensions.

6 NUMERICAL RESULTS

This section evaluates COLA on the problem in equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is the counterfactual returned by a CE method A_{CE} . To probe complementary aspects of distributional discrepancy, we report results under four divergence functions, each sensitive to a different facet of the discrepancy: OT measures the distance between entire distributions; MMD measures the discrepancy between distribution means in a high-dimensional feature space; and absolute mean difference (MeanD) and absolute median difference (MedianD) measure the discrepancy in mean and median outcome, respectively. The experiments are organized to substantiate three claims, each isolated by a dedicated, controlled comparison: I) *COLA achieves substantial modification minimality.*

II) p -SHAP outperforms competing Shapley methods on modification minimality. III) COLA attains near-optimal performance on tractable benchmarks.

Experiment Setup We evaluate across a deliberately broad design: 4 binary-classification datasets, 5 CE algorithms spanning diverse goals, and 12 classifiers, all enumerated in Table 1. Each combination of a dataset, an A_{CE} algorithm, and a model f constitutes an “experiment scenario,” so that the full design spans a large and heterogeneous scenario space—encompassing linear, kernel, tree-ensemble, and deep models alongside instance-level, group-level, and distributional generators—over which COLA is systematically benchmarked. This breadth is precisely what substantiates the general, model- and generator-agnostic claims advanced throughout the paper.

Table 1: [Experiment Scenarios Setup] Four datasets serve to benchmark five CE algorithms across 12 models spanning linear, kernel, tree-ensemble, and deep architectures. A “scenario” is any combination of a dataset, an A_{CE} algorithm, and a model f ; jointly, these combinations define the full space over which the methods are compared.

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The 6 compared methods, color-coded to match Figure 3. Each method is a specific instantiation of the triple $(A_{\text{Prob}}, A_{\text{Value}}, A_{\text{Shap}})$ and is benchmarked within the experiment scenarios of Table 1. CF- p_{OT} is the proposed p -SHAP; the remaining five are baselines and ablations, deliberately constructed so that successive rows toggle a single design choice and thereby isolate its individual effect.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

The five A_{CE} generators in Table 1 were selected expressly to exercise the full spectrum of CE paradigms. DiCE and KNN are instance-based methods that produce counterfactual(s) for each individual factual instance; AReS and GlobeCE are group-based methods that return a collection of counterfactuals for the factual data as a whole; and Discount treats the factual instances as an empirical distribution and seeks a proximate counterfactual distribution. This deliberate coverage enables us to assess COLA against instance-level, group-level, and distributional CE within a single, uniform evaluation protocol, so that the resulting conclusions extend across the entire CE taxonomy rather than any one paradigm.

Table 2 specifies the 6 compared methods, of which CF- p_{OT} is the proposed p -SHAP and the remaining five are baselines. Each method is fully determined by the triple $(A_{\text{Prob}}, A_{\text{Value}}, A_{\text{Shap}})$ and is benchmarked comprehensively across the scenarios of Table 1. To take the first row as an example, RB- p_{Uni} sets A_{Prob} to the uniform distribution to obtain $\mathbf{p}$, feeds $\mathbf{p}$ into A_{Value} via equation (9) or equation (10) to compute $\mathbf{q}$, and adopts RB-SHAP as A_{Shap} .

These six methods constitute a controlled ablation engineered so that each pairwise comparison isolates exactly one design choice. First, RB- p_{Uni} and CF- p_{Uni} differ only in whether CE information enters the attribution, thereby isolating the value of using CE at all. Second, CF- p_{Uni} , CF- p_{Rnd} , and

CF- p_{OT} share an identical use of CE in FA but differ in the joint distribution employed, thereby isolating the contribution of the coupling itself—and, as we demonstrate below, the OT coupling markedly outperforms both the uniform and the random alternatives. Third, to confirm that OT helps because it supplies A_{Shap} with a genuine factual–counterfactual alignment rather than through some incidental effect, we compare CF- p_{OT} , which exploits this alignment, against RB- p_{OT} , which does not, holding all else fixed. Finally, CF- p_{Ect} instantiates the idealized special case in which each counterfactual originates from a known source, furnishing an exact factual–counterfactual alignment and collapsing CF-SHAP onto B-SHAP. Taken together, the six methods trace a controlled progression from alignment-free to fully alignment-informed attribution, so that the empirical gains can be attributed unambiguously to the coupling.

Table 3: [COLA for Modification Minimality] For each method, the number of modified features in $\mathbf{z}$ required for $f(\mathbf{z})$ to reach 80% (and 100%) of the counterfactual effect of $f(\mathbf{r})$; the adjacent column reports the relative displacement $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$, so that lower values in both columns are better. Each entry averages 4 randomly selected scenarios from Table 1 with 100 runs per scenario, and boldface marks the best (smallest) value within each block. The symbol “—” indicates that the target counterfactual effect is unattainable by that method. The pattern is unambiguous: the proposed CF- p_{OT} attains the best value in every block and is the sole method to reach the demanding 100% effect on all four datasets, achieving it with as little as 26–45% of the displacement incurred by the original CE.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z} - \mathbf{x}\ }{\ \mathbf{r} - \mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z} - \mathbf{x}\ }{\ \mathbf{r} - \mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

Result I: COLA delivers substantial action reduction at negligible loss of counterfactual effect

In Table 3, we require $\mathbf{z}$ to reach 80% or 100% of the counterfactual effect of $f(\mathbf{r})$ ². The evidence is decisive: COLA attains these targets while modifying substantially fewer features of $\mathbf{z}$ than the original CE $\mathbf{r}$. To recover 80% of the counterfactual effect, COLA with p -SHAP traverses only 13–25% of the distance of the original CE (measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$); and, as Table 3 shows, it is the only method able to reach the full 100% effect on every dataset, doing so at merely 26–45% of the edits made by the original CE $\mathbf{r}$. p -SHAP thereby delivers the strongest action minimality among all six methods considered—a margin that holds uniformly across every dataset and that we dissect in detail below.

²By definition $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which corresponds to 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. A counterfactual effect of 80% is defined via the fraction of divergence eliminated by the refined CE $\mathbf{z}$, i.e. $Counterfactual\ Effect = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D taken to be OT.

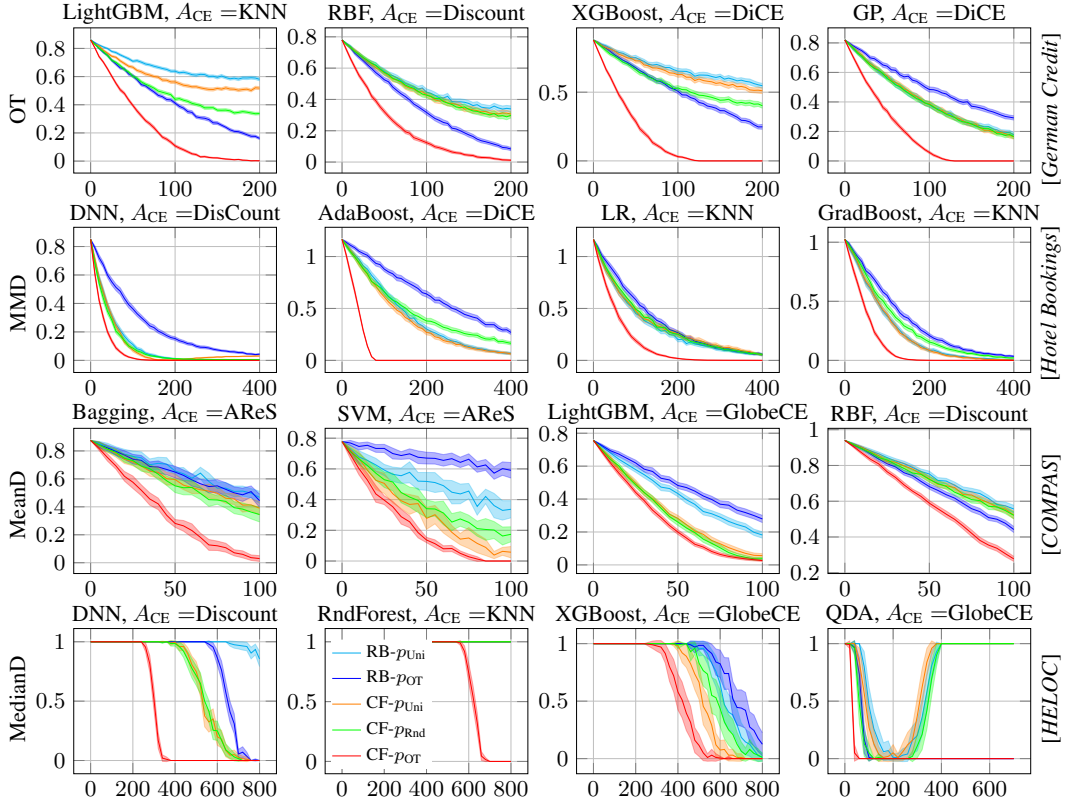


Figure 3: Residual counterfactual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C ; lower is better. Each curve averages 100 runs, with shaded bands showing 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, and $A_{\text{Value}}^{\text{max}}$ for *German Credit* and *Hotel Bookings*. The legend in the “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” panel applies to all plots. Across every panel—spanning all four datasets and diverse model–generator pairs—the proposed $\text{CF-}p_{\text{OT}}$ (red) attains the lowest divergence for essentially every budget C , and its advantage widens most sharply at small C , precisely the regime in which sparse action plans are most valuable.

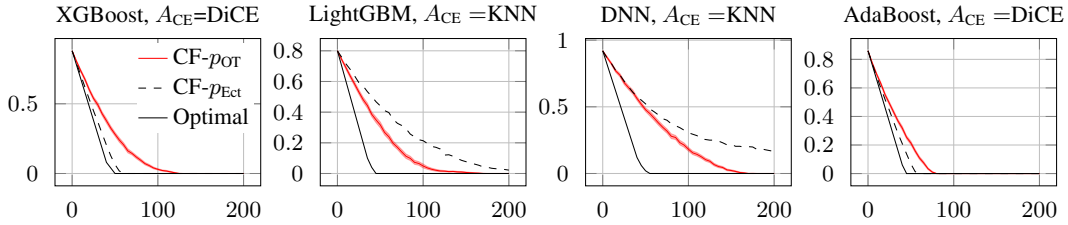


Figure 4: [*German Credit*] Residual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ against the action budget C , with $D = \text{MeanD}$, comparing $\text{CF-}p_{\text{OT}}$ and $\text{CF-}p_{\text{Ect}}$ against the mixed integer linear programming (MILP) global optimum. The proposed $\text{CF-}p_{\text{OT}}$ tracks the optimum closely and, in several cases, matches or even surpasses the alignment-informed $\text{CF-}p_{\text{Ect}}$ —and it does so without ever observing the true factual–counterfactual alignment that $\text{CF-}p_{\text{Ect}}$ is granted, underscoring that an inferred OT coupling can rival an exact one.

Result II: p -SHAP outperforms competing Shapley methods in achieving the counterfactual effect Figure 3 evaluates the Shapley variants of equation (4)–equation (7), plotting the residual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1) (vertical axis) against the number of allowed feature changes C (horizontal axis). Three findings emerge, each following directly from a single controlled contrast in Table 2. First, $\text{RB-}p_{\text{Uni}}$ and $\text{RB-}p_{\text{OT}}$ trail all other methods by a wide margin, which establishes that exploiting CE information is essential for attributing features towards modification

minimality. Second, because $\text{CF-}p_{\text{OT}}$ and $\text{RB-}p_{\text{OT}}$ differ only in A_{Shap} , the dominance of $\text{CF-}p_{\text{OT}}$ over $\text{RB-}p_{\text{OT}}$ pinpoints the source of the gain unambiguously: OT helps precisely because it supplies an effective factual–counterfactual alignment, and not through any confound. Third, p -SHAP substantially outperforms both $\text{CF-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Rnd}}$, which demonstrates the decisive role of the OT coupling within p -SHAP. In sum, merely injecting counterfactual information into FA (as $\text{CF-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Rnd}}$ do) is insufficient; what proves decisive is a *proper* factual–counterfactual alignment—and, as Figure 4 subsequently reveals, this alignment need not coincide with the one dictated by the exact counterfactual generation mechanism.

COLA with p -SHAP performs strongly and consistently across the entire scenario space of Table 1, as Figure 3 and Table 3 jointly attest, and an extensive further body of corroborating results is reported in Appendix H. Taken together, these experiments establish that p -SHAP is the uniformly strongest among the competing Shapley methods for inducing sparsity in CE.

Result III: COLA is competitive with the MILP optimum on tractable benchmarks This result demonstrates the effectiveness of p -SHAP in eliminating the influence of the CE generation process by replacing the CE algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT joint distribution between the factual and counterfactual data, shown in Figure 4. We benchmark $\text{CF-}p_{\text{Ect}}$ within COLA, solving equation (1) with MeanD as the divergence D . Because $\text{CF-}p_{\text{Ect}}$ presupposes a known factual–counterfactual alignment, this experiment quantifies precisely how effectively COLA exploits such an alignment when one is available. The corresponding theoretical optimum is obtained by solving an MILP, whose derivation appears in Appendix G; because the MILP is computationally heavy, we compute it only for *German Credit*, which is small enough to remain tractable, thereby furnishing a genuine global-optimality yardstick against which the approximate methods can be calibrated.

Figure 4 shows that $\text{CF-}p_{\text{Ect}}$ attains near-optimal performance under DiCE. Since the factual–counterfactual pairs produced by DiCE and KNN are mutually independent, this confirms that COLA is effective for instance-based CE—a conclusion the evidence fully supports, and one that holds even though the formulation in equation (1) is a strict generalization further covering group and distributional CE. Most striking of all, $\text{CF-}p_{\text{OT}}$ at times surpasses $\text{CF-}p_{\text{Ect}}$, because it utilizes a more theoretically grounded approach to identify the key features that require modification, whereas $\text{CF-}p_{\text{Ect}}$ relies on CE algorithm-dependent knowledge, which lacks solid justification on its effectiveness for FA: an inferred OT coupling can constitute a better attribution reference than the generator’s own exact alignment. A residual gap between $\text{CF-}p_{\text{Ect}}$, $\text{CF-}p_{\text{OT}}$, and the exact optimum persists under KNN, so characterizing the best attainable alignment stands out as a well-motivated open question and a natural avenue for further gains.

7 CONCLUSIONS

We introduced COLA, a model- and generator-agnostic framework that refines counterfactual explanations through joint-distribution-informed Shapley values (p -SHAP), preserving the counterfactual effect while requiring markedly fewer actions. The central idea is to employ an OT coupling to endow Shapley attribution with a proper factual–counterfactual alignment—an idea that unifies the existing Shapley variants as special cases of a single scheme and that, under a mild Lipschitz condition, provably minimizes an upper bound on the W_1 divergence while keeping the refined counterfactual no farther from the factual than the original. Empirically, across four datasets, twelve models, and five CE generators spanning instance-level, group-level, and distributional counterfactuals, COLA with p -SHAP attains the target effects with only 26%–45% of the original edits and closely tracks the global optimum on a tractable benchmark. These results establish COLA as a general and reliable route to sparse, actionable counterfactuals; and because the coupling algorithm A_{Prob} enters as a modular component, the design of still stronger couplings—the one locus where both the analysis and the experiments indicate room to close the residual optimality gap—stands out as a concrete and promising direction for future work.

³This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a method (*COLA* with *p-SHAP*) for refining CE. The work is methodological and empirical in nature: it involves neither human subjects nor interventions, and we neither collect, annotate, nor release any new personal data.

All experiments rely on standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We adhere both to the applicable dataset licenses and terms and to the established research practice of using these data solely for benchmarking; we attempt neither re-identification nor linkage, and we release no additional personal information. Our code will expose options to exclude protected attributes and to enforce feasibility and immutability constraints on features.

CE can surface biases latent in the underlying models, yet they do not by themselves guarantee fairness. Our method reduces the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age), nor does it confer causal validity. We accordingly discourage the use of CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains should incorporate domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact or benefit across groups), human oversight, and compliance with the applicable data-protection and AI regulations. We stress that our attribution is distributional rather than causal, and hence must not be interpreted as a causal effect.

The principal risks are threefold: (i) misinterpreting attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate them, respectively, by (a) stating the non-causal scope and limitations explicitly in the paper, (b) supporting cost/feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) providing guidance for responsible use in the documentation.

To facilitate replication, we provide an anonymized implementation together with detailed settings and seeds; we report model types, hyperparameters, and compute budgets, and we guard against cherry-picking through fixed evaluation protocols. To our knowledge, this submission entails no conflicts of interest or sensitive sponsorships; any funding disclosures will be provided after the double-blind review.

REPRODUCIBILITY STATEMENT

We treat reproducibility as a first-class concern and supply pointers to every component required to reproduce our results. The COLA framework and *p-SHAP* attribution are fully specified through pseudocode and notation in the main paper (Algorithm 1, together with the Method and Preliminaries sections), while all theoretical assumptions and complete, line-by-line proofs are provided in the appendix.

Implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which further describes the datasets and the evaluation protocol, including data splits and preprocessing. The baselines (models and CE generators) and coupling choices are enumerated exhaustively in the experimental sections and ablations, and their configurations are mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; Appendix I additionally reports hardware and runtime to contextualize the compute requirements, and the appendix furnishes ablation and sensitivity studies together with the small-scale MILP benchmark setup, so that every reported claim can be independently verified.

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APPENDICES

A	Comparison to Existing Approaches	16
B	$\mathcal{NP}$-hardness of the Problem in equation (1)	16
C	Proof of Theorem 4.1: p-SHAP Towards Counterfactual Effect	17
D	Proof of Theorem 4.2: Interventional Effect	19
E	Proof of Theorem 5.1: Counterfactual Proximity	20
F	Analysis of Computational Complexity of COLA	21
G	An MILP Formulation of equation (1) With MeanD	21
H	Extended Numerical Results	22
I	Experiments Reproducibility	23

A COMPARISON TO EXISTING APPROACHES

Albini et al. (2022) proposed CF-SHAP, which employs counterfactual data points as the background distribution for Shapley attribution. Its formulation, however, conditions the counterfactual data distribution on each individual instance, thereby presupposing a known probabilistic alignment between every factual and every counterfactual instance; comparable assumptions underlie (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). This assumption breaks down in the many settings where global explanations are sought, whereas our framework subsumes the counterfactual-distribution setup of (Albini et al., 2022) as a single special case obtained by a particular choice of A_{Prob} . Kwon & Zou (2022) proposed WeightedSHAP, which weights features rather than treating them as equal contributors; by contrast, our method weights the contributions of data points and generalizes naturally to weighting both rows and columns. Kommiya Mothilal et al. (2021) establish a framework that jointly leverages FA and CE for explainability, yet their CE-based FA inherits the same alignment assumption as (Albini et al., 2022), which obstructs generalization to the group (Ley et al., 2023; 2022) and distributional CE (You et al., 2025) settings that our formulation handles uniformly. A further related line, (Sharma et al.), models counterfactuals through a system’s causal structure and time-series predictors in order to attribute an observed change in a system-level metric to individual inputs. Crucially, none of the foregoing work addresses the minimal-action CE problem that is the focus of this paper, which distinguishes our contribution from all of these lines.

The problems formulated in (Kanamori et al., 2022; Karimi et al., 2021) are the closest to ours, and contrasting them sharpens our contribution. Karimi et al. (2021) minimize the cost of the actions performed but assume a known structural causal model (SCM), which is seldom available in practice. Kanamori et al. (2022) observe that it remains open whether existing CE methods can solve their problem, and propose an MILP-solver-based approach for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Solving the MILP is nonetheless costly and scales poorly—exactly the limitation that our post-hoc, model-agnostic framework is engineered to circumvent, trading exact combinatorial optimization for a provably cost-bounded, polynomial-time refinement.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 below establishes that no scalable exact algorithm can be expected for the general problem unless $\mathcal{P} = \mathcal{NP}$, and, more pointedly, that the hardness stems not from any non-linearity of f but from the combinatorial structure of the action-selection constraint itself—which is precisely why it persists even in the simplest linear, one-dimensional regime.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. The argument proceeds by reduction from a canonical $\mathcal{NP}$ -hard problem. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix

$\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the objective is to determine a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error subject to retaining at most K non-zero entries:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero entries of $\mathbf{z}$. This problem is $\mathcal{NP}$ -hard precisely because of the combinatorial choice of which variables to include in the model.

We reduce this SR problem to our problem equation (1) through the following construction. Fix the number of features at $d = 1$ and the number of instances at n (matching the dimension of the SR problem); let the factual data be the zero vector $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$ and the target output be $\mathbf{y}^* \in \mathbb{R}^m$ as given, and take the model f to be the linear map $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the output divergence we set $D(f(\mathbf{z}), \mathbf{y}^*) = \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2$, the squared Euclidean distance, and for the instance divergence $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, since $\mathbf{x} = \mathbf{0}$. We equate the action budget C with the sparsity level k of the SR problem, and take the large constant M to be any sufficiently large positive number, e.g. $M \geq \max_i |z_i|$.

Because $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) simplify to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

Hence, if $c_i = 0$ the two bounds force $z_i = 0$, whereas if $c_i = 1$ then $z_i \in [-M, M]$, and constraint equation (1c) reduces to $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thereby takes the form:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation, each binary variable c_i encodes whether z_i is permitted to change ($c_i = 1$) or is held fixed ($c_i = 0$). The constraints pin $z_i = 0$ whenever $c_i = 0$, thereby mirroring exactly the sparsity constraint of the SR problem, while $\sum_{k=1}^n c_i \leq k$ caps the number of changeable features at k to match the sparsity level; the objective, in turn, coincides with that of SR.

Since this construction reproduces the SR problem with a cardinality constraint—itsself $\mathcal{NP}$ -hard—solving Problem equation (1) is at least as hard as solving SR. Consequently, Problem equation (1) is $\mathcal{NP}$ -hard even in the maximally restricted regime in which $d = 1$, f is linear, and the divergences D are standard Euclidean distances—which confirms that the intractability is intrinsic to the action-selection combinatorics rather than an artifact of model complexity. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ denote the factual data points, carrying probability weights $\mu_i \geq 0$ with $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ denote the counterfactual data points, carrying weights $\nu_j \geq 0$ with $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below makes precise the sense in which the feature

attributions align with the expected costs of feature modification, and thereby guarantees that the induced action plans are cost-efficient in attaining the counterfactual outcome.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$ we have

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides by $p_{ij}^{OT} \geq 0$ yields

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

and summing over all $i = 1, \dots, n$ and $j = 1, \dots, m$ gives

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Writing $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, the inequality reads

$$E_f \leq L E_d.$$

To bound E_d in turn, we invoke the Cauchy–Schwarz inequality, noting that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$, since p_{OT} is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. For any real-valued arrays a_{ij} and b_{ij} , Cauchy–Schwarz gives

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Choosing $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we obtain

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying Cauchy–Schwarz,

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this reduces to

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

and taking square roots yields

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the bound on E_f gives

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Observe that the 1-Wasserstein divergence is no larger than E_f^5 , whence

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Moreover, since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$,

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

This establishes the claimed bound and completes the proof. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ denote the joint probability of $\mathbf{x}$ and $\mathbf{r}$ returned by A_{prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are clamped to $\mathbf{x}_{i,S}$ while those in $\mathcal{F} \setminus S$ remain distributed according to their marginal $p(\mathbf{r}_{\mathcal{F} \setminus S})$. The intervened expectation of $f(\mathbf{r})$ is therefore

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

By the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

which completes the proof. $\square$

The value function $v^{(i)}(S)$ thus captures precisely the expected value of $f(\mathbf{r})$ under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$ that clamps the features in S to $\mathbf{x}_{i,S}$. Because this intervention is independent of any predefined joint distribution $p(\mathbf{x}, \mathbf{r})$, the quantity $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ decomposes cleanly into the base expected value of $f(\mathbf{r})$ and the incremental interventional contribution of the attribution $v^{(i)}(S)$, which endows the attribution with a well-defined operational meaning rather than a merely correlational one.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Although both π and $\mathbf{p}_{\text{OT}}$ are joint distributions of $\mathbf{x}$ and $\mathbf{r}$, π minimizes the W_1 summation over all admissible joint distributions whereas $\mathbf{p}_{\text{OT}}$ need not, so $W_1 \leq E_f$.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. At every modified entry ($c_{ik} = 1$) we have $z_{ik} = q_{ik}$, while at every unmodified entry ($c_{ik} = 0$) we have $z_{ik} = x_{ik}$. Consequently,

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms relative to $\mathbf{q}$ and $\mathbf{z}$ are

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where equality (i) follows from equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, we conclude that

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

and taking square roots yields

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

which completes the proof. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ is a permutation matrix induced by a bijection σ . Consequently, the value-assignment algorithm A_{value} —whether maximizing or averaging—assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting the permuted matrix $\mathbf{r}_\sigma$ for $\mathbf{q}$ in the general bound of Theorem 5.1 (namely $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) yields the claim immediately. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We analyze the computational complexity of COLA component by component, and then aggregate the resulting terms.

First consider A_{Prob} . When the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, A_{Prob} merely materializes the matrix $\mathbf{p}$ from that prior knowledge, at cost $\mathcal{O}(n \times m)$. Otherwise, obtaining $\mathbf{p}$ by solving OT via the Sinkhorn–Knopp algorithm costs $\mathcal{O}(n \times m \times \log(1/\varepsilon))$, so that the regularization parameter ε enters the complexity only logarithmically.

Next consider A_{Shap} . For each sampled subset, the model is evaluated over all n data points, yielding $\mathcal{O}(n)$ evaluations per subset. Incorporating baseline values from the reference data $\mathbf{r}$ entails substituting the selected features with their reference counterparts; this operation reads the baseline value from the reference table $\mathbf{r}$ for each of the d features and costs $\mathcal{O}(d)$, provided the reference values are precomputed and accessible in constant time. With M_{Shap} sampled subsets, and hence M_{Shap} rounds of model evaluation, the total complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA costs $\mathcal{O}(n \times d)$.

We now compare the two value assigners, $A_{\text{Value}}^{\max}$ and $A_{\text{Value}}^{\text{avg}}$, step by step, since they incur different costs. For $A_{\text{Value}}^{\max}$, each row $\mathbf{x}_i$ requires (1) computing the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, in $\mathcal{O}(m)$ operations; (2) identifying the reference row $\mathbf{r}_j$ of highest probability p_{ij} , again in $\mathcal{O}(m)$; and (3) assigning $q_{ik} = r_{\tau(i),k}$ with $\tau(i) = \arg \max_j p_{ij}$, in $\mathcal{O}(d)$. Over all n rows, the total complexity is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, each row $\mathbf{x}_i$ requires (1) computing the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, in $\mathcal{O}(m)$; (2) computing the normalizer $\sum_{j'=1}^m p_{ij'}$, in $\mathcal{O}(m)$; and (3) forming the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, in $\mathcal{O}(m \times d)$. Over all n rows, the total complexity is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the complexity is $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$, since $C \leq n \times d$.

Aggregating these terms, the complexity of COLA with $A_{\text{Value}}^{\max}$ is

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA with $A_{\text{Value}}^{\text{avg}}$ is

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence, in terms of n , m , d , and the entropic-OT regularization parameter ε , the complexity of COLA is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ when $A_{\text{Value}}^{\max}$ is used and $N = \mathcal{O}(nmd)$ when $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This section derives the global-optimality benchmark employed in Figure 4 for the case of a known factual–counterfactual alignment, obtained by solving equation (1) with D set to MeanD, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

where $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Because COLA guarantees $D(\mathbf{r}, \mathbf{x}) \leq \varepsilon$ and, by Theorem 5.1, keeps $\mathbf{z}$ at least as close to $\mathbf{x}$ as $\mathbf{r}$ is, the proximity constraint equation (1b) is automatically satisfied and may

therefore be dropped. Problem equation (1) accordingly reduces to

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

The original constraints equation (1d) and equation (1e) collapse into the single constraint equation (13c), precisely because CF- p_{Ect} is imposed: each $\mathbf{x}_i$ possesses an exact reference $\mathbf{r}_j$ used by both A_{Shap} and A_{Value} , so that every x_{ik} ($k = 1, 2, \dots, d$) is either held fixed or set to r_{jk} . The value $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ is therefore governed entirely by the binary variable c_{ik} .

Under this known alignment between each $\mathbf{x}_i$ and its $\mathbf{r}_j$, the matrix $\mathbf{q}$ is fully determined, and $A_{\text{Values}}^{\text{max}}$ and $A_{\text{Values}}^{\text{avg}}$ return an identical $\mathbf{q}$. For any instance i and any feature subset $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denote the solution $\mathbf{z}_i$ in which every feature $k \in S$ is set to q_{ik} and every remaining feature $h \in \mathcal{F} \setminus S$ retains x_{ih} ; the collection $\{\mathbf{z}_{iS}\}_{S \subseteq \mathcal{F}}$ then enumerates all admissible values of $\mathbf{z}_i$. Define a scalar variable for each $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ is represented exactly by a binary selector a_{iS} together with the scalar g_{iS} :

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is therefore reformulated as the mixed-integer program below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η subject to equation (14b) and equation (14c) is equivalent to minimizing the objective of equation (13), since together these two constraints bound $|\sum_{i,S} g_{iS} a_{iS}|$ from above by η . Constraint equation (14d) enforces that each instance i selects exactly one modification plan a_{iS} for some $S \subseteq \mathcal{F}$, whereas equation (14e) implements the action budget equation (13b). Solving equation (14) therefore yields the exact global optimum of COLA under a known factual-counterfactual alignment—the benchmark against which CF- p_{OT} and CF- p_{Ect} are measured in Figure 4 of Section 6.

H EXTENDED NUMERICAL RESULTS

We now analyze how the choice of Shapley method affects FA, corresponding to lines 3–4 of COLA, through two controlled comparisons that each vary only A_{Shap} . First, RB- p_{Uni} and CF- p_{Uni} differ solely in A_{Shap} , and across Figures 5–8 CF- p_{Uni} consistently outperforms RB- p_{Uni} . Second, RB- p_{OT} and CF- p_{OT} likewise differ only in A_{Shap} , and the latter uniformly outperforms the former. Because each comparison isolates A_{Shap} alone, together they establish decisively that *RB-SHAP is unsuitable for FA in CE*.

A second and equally consequential step of COLA is line 5, which uses the joint probability $p(\mathbf{x}, \mathbf{r})$ to compose the matrix $\mathbf{q}$ and thereby prescribes the direction in which each factual x should move to approach the target model outcome. Across Figures 5–8, CF- p_{OT} outperforms every other method in every experiment. Although CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all supply a joint probability $\mathbf{p}$ in the regime where the exact factual–counterfactual alignment is unknown, their performance diverges sharply—which confirms that merely knowing the CE (together with its marginal distribution) is insufficient, and that the quality of the coupling is the decisive factor.

OT proves especially valuable precisely when alignment information between factual and counterfactual instances is missing or unreliable. Indeed, even when the CE algorithm explicitly matches each factual instance to a counterfactual, there is no guarantee that this known alignment maximizes performance—a point corroborated by Figure 4 in Section 6, in which the inferred OT coupling occasionally outperforms the exact one.

Crucially, p_{OT} need not coincide with the true data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$; its role is instead to guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ jointly for both FA and CE. The QDA column of Figure 5 further illustrates the stability of the OT-based methods, whereas the alternatives diverge markedly from the target. We emphasize, however, that COLA is *not restricted to OT* as A_{Prob} : as Figure 4 indicates, even the best known $\mathbf{p}$ leaves a non-negligible gap to the global optimum, so that designing an improved A_{Prob} remains a concrete and promising avenue for future work.

I EXPERIMENTS REPRODUCIBILITY

The experiments are executed on a high-performance computing (HPC) cluster of four nodes (one per dataset) operating in parallel; each node is equipped with two Intel Xeon 2660v3 processors (10 cores, 2.60 GHz) and 128 GB of memory. Each node completes its workload in roughly 5–10 hours, depending on dataset size. The experiments can equally be reproduced on a laptop, albeit at a generally higher computational cost than on the HPC cluster.

For all four datasets, numerical features are standardized and categorical features are encoded by either label encoding or one-hot encoding; in practice we observed no appreciable difference between the two encodings in COLA’s performance, which indicates that the framework is robust to this preprocessing choice. The train–test split is fixed at 7 : 3 throughout.

The optimality baseline in Figure 4 is solved with Gurobi 11.0.2 ([gur](#)), which requires a Gurobi license; in its absence, open-source operations-research libraries such as Google OR-Tools ([goo](#)) may be substituted. Because solving the MILP of Appendix G is computationally expensive, it remains practical only for small-scale datasets such as German Credit; computing the optimality baseline for larger datasets would require reducing the number of features considered.

The model hyperparameters are as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are scikit-learn models ([skl](#)) with all hyperparameters left at their defaults. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)): the DNN comprises three layers, and the SVM employs a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used through their scikit-learn interfaces, again with default hyperparameters.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) solely as general-purpose assistive tools, and only for grammar, wording, and light copy-editing of author-written text. LLMs contributed nothing to research ideation, modeling choices, experimental design, or results: all algorithms, proofs, datasets, and analyses were conceived and verified exclusively by the authors. Every LLM-suggested phrasing was reviewed and edited prior to inclusion. The authors take full responsibility for all content; LLMs are neither authors nor contributors. This disclosure complies with the ICLR policy on LLM usage.

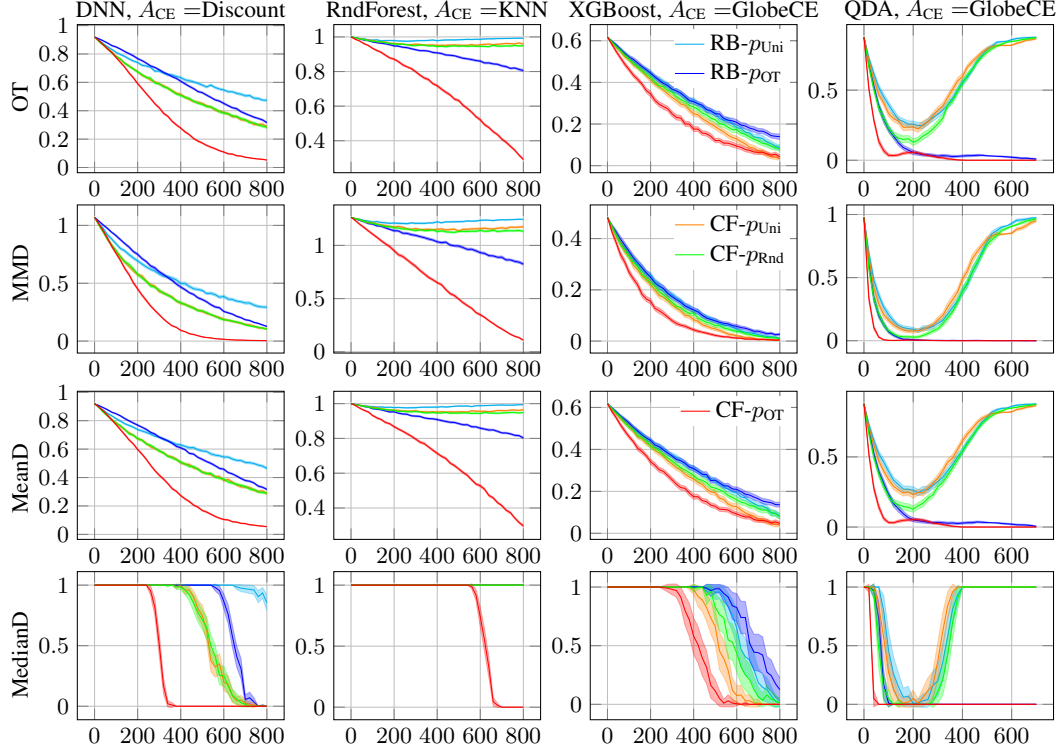


Figure 5: [HELOC] Residual counterfactual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C ; lower is better. Each curve averages 100 runs, with shaded bands showing 99.9% confidence intervals, and the legend applies to all panels. Here $A_{\text{Value}}^{\text{avg}}$ is used. Across every model and every divergence measure, CF- p_{OT} attains the lowest divergence, confirming that its advantage is not confined to any particular model class or metric.

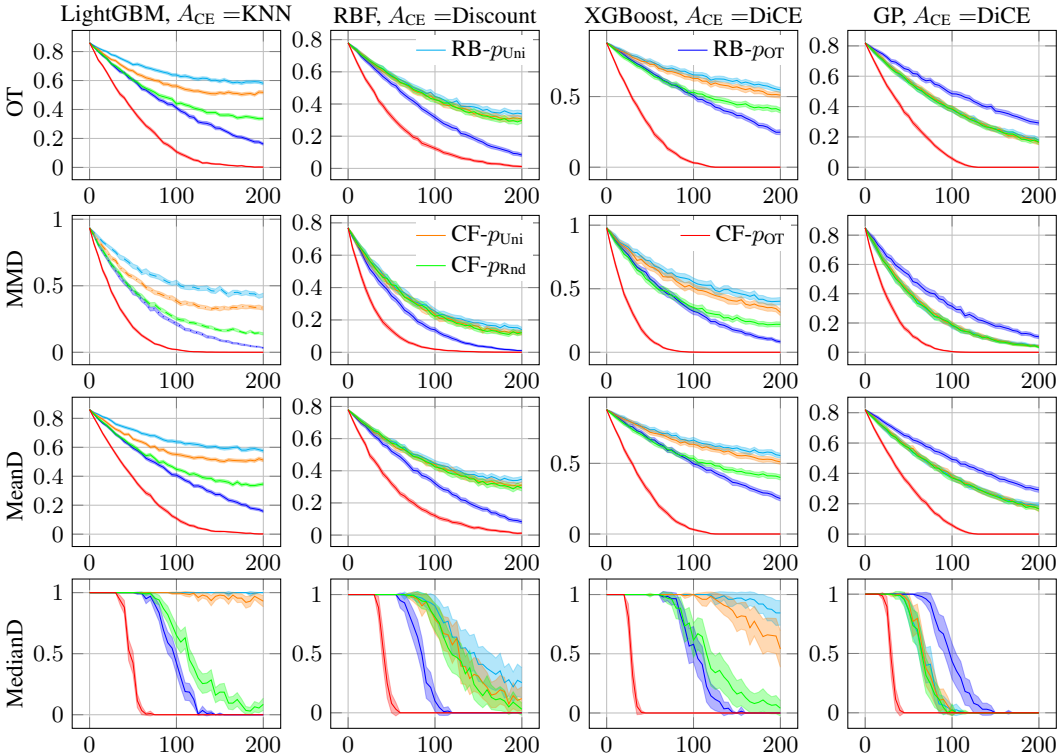


Figure 6: [German Credit] Residual counterfactual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C ; lower is better. Each curve averages 100 runs, with shaded bands showing 99.9% confidence intervals, and the legend applies to all panels. Here $A_{\text{Value}}^{\text{avg}}$ is used. Across every model and every divergence measure, CF- p_{OT} attains the lowest divergence, extending the consistency observed on the other datasets to German Credit as well.

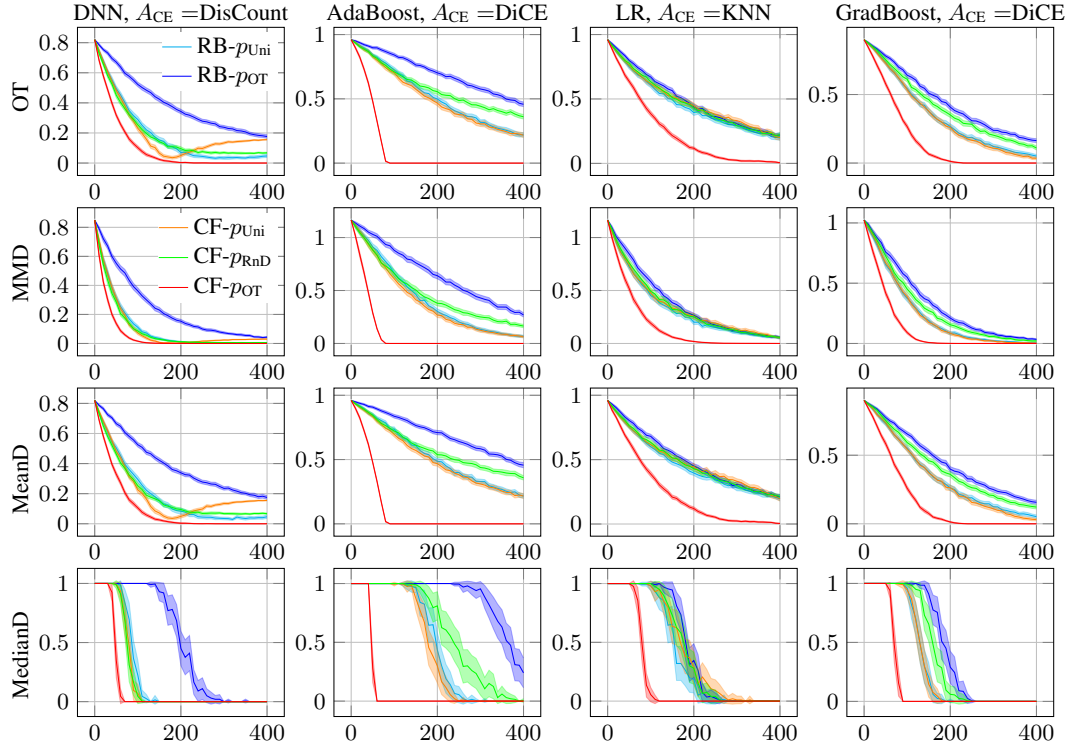


Figure 7: [*Hotel Bookings*] Residual counterfactual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C ; lower is better. Each curve averages 100 runs, with shaded bands showing 99.9% confidence intervals, and the legend applies to all panels. Here $A_{\text{Value}}^{\max}$ is used. Across every model and every divergence measure, CF- p_{OT} attains the lowest divergence, and the same dominance holds under the single-reference assigner $A_{\text{Value}}^{\max}$ used here.

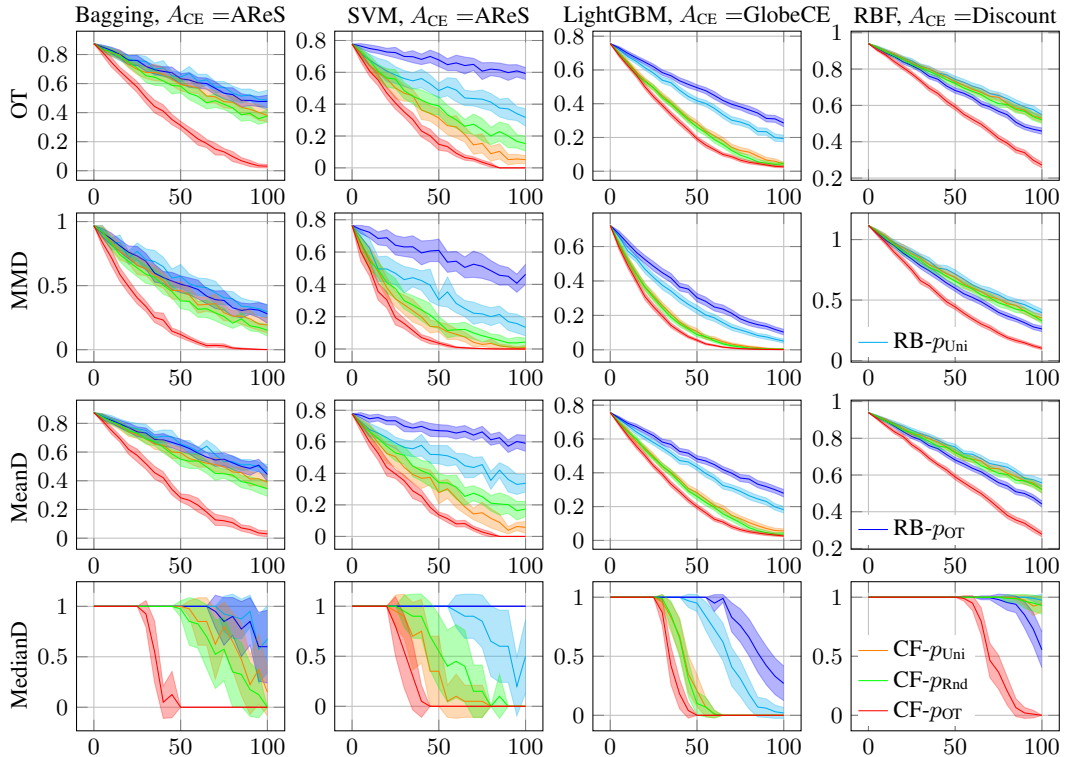


Figure 8: [*COMPAS*] Residual counterfactual divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C ; lower is better. Each curve averages 100 runs, with shaded bands showing 99.9% confidence intervals, and the legend applies to all panels. Here $A_{\text{Value}}^{\max}$ is used. Across every model and every divergence measure, CF- p_{OT} attains the lowest divergence, completing the uniform pattern of dominance observed across all four datasets.